merman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL https://transformer-circuits.pub/2025/attribution-graphs/biology.html.

Max M Louwerse and Nick Benesh. Representing spatial structure through maps and language: Lord of the rings encodes the spatial structure of middle earth. *Cognitive science*, 36(8):1556–1569, 2012.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt, 2023. URL https://arxiv.org/abs/2202.05262.

Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. Talking heads: Understanding inter-layer communication in transformer language models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL https://openreview.net/forum?id=LUsx0chTsL.

Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26, 2013.

nostalgebraist. Interpreting GPT: The logit lens. *LessWrong*, 2020. URL https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

Andrei Popescu-Belis, Àlex R. Atrio, Bastien Bernath, Etienne Boisson, Teo Ferrari, Xavier Theimer-Lienhard, and Giorgos Vernikos. GPoeT: a language model trained for rhyme generation on synthetic data. In Stefania Degaetano-Ortlieb, Anna Kazantseva, Nils Reiter, and Stan Szpakowicz, editors, *Proceedings of the 7th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature*, pages 10–20, Dubrovnik, Croatia, May 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.latechclfl-1.2. URL https://aclanthology.org/2023.latechclfl-1.2/.

A. Emergence of third-party vowels in interventions

In the Figure 2 example, the vowel sounds remain relatively isolated. Sometimes, however, unexpected third-party vowel sounds appear on the fringes, such as small /o/-to-/ɛ/ interventions manifesting /i/ sounds, as shown in Figure 6.

These “intermediate" vowel sounds often appear in consistent patterns for consistent ξ and μ vowels, regardless of

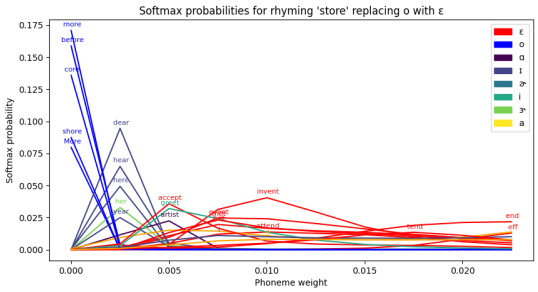

Figure 6: Example intervention on rhymes with store. When replacing the /o/ sound with /ɛ/ at small phoneme weight values, the model predicts words containing /i/. This is consistent across multiple starting words containing /o/.

the consonants involved in the starting word, which suggests that the model internally represents vowel sounds in the embedding space with meaningful relationships to each other. This motivates our attention to colinearities of phoneme vectors in PCA space in Section 4. When applying this methodology to non-phonetic interpretability tasks, we suggest carefully studying the presence of third-party characteristics (i.e. attributes of model results which are not strictly related to the specific intervention) in order to inform both the choice of dimensionality reduction technique and the expected emergent geometric patterns in PCA space.

B. Activation patching results

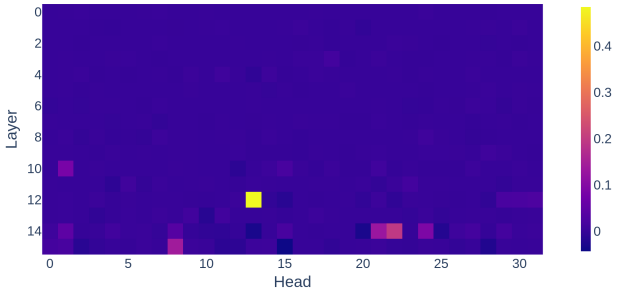

Figure 7: The mean normalized logit differences for our activation patching experiment described in Section 3. Head 13 Layer 12 clearly has the highest value at 0.48.

C. Exploring the head dimension

After discovering H13L12, we were interested in understanding both how the internal head dimension is organized (e.g. are there phoneme specific neurons in the output matrix?) and how phonetic information is pulled from the target word into the head.

The model seems to use complex interference patterns to

approximate different phonetic directions. This is evidenced by the relative uninterpretability of the head dimension and the result matrix combined with the mutual consistency of similar phonemes within the head dimension with respect to cosine similarity. Notably, we were able to recover much of the result vector (measuring with cosine similarity) with the top and bottom (In terms of magnitude) eight values from the head vector, suggesting a kind of sparsity in the head dimension (See Figure 8). Interestingly, we found that all sixty four head dimensions were, for at least one word, present within the top or bottom eight z values. Future work could investigate the subspaces read from and written to by this head using more involved methods, like sparse coding (Kissane et al.) or communication channels (Elhage et al., 2021; Merullo et al., 2024).

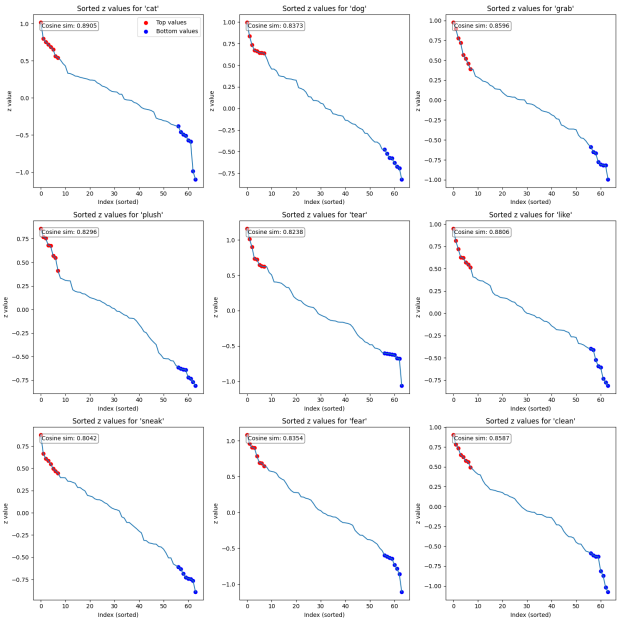

Figure 8: Sorted Z values with the top and bottom eight highlighted. Cosine similarity is between the full result vector and the vector generated by only taking the top and bottom eight values (setting every other index to zero) and passing it through the output matrix.

D. Manual inspection of decoded result vectors

To better understand the role the phoneme mover head plays in downstream rhyme production, we perform a manual survey of one hundred different runs of our rhyme task (described in Section 2), chosen randomly after filtering for tokenization from the Oxford 5000 dataset (Güzel, 2025).

We judge result vectors according to the following criteria: we call a result vector (R.V.) "*coherent*" if and only if five of the top ten tokens it promotes contain similar phonemes to the target rhyme. For example, if the target word was `plush` then the target rhyme would be /ʌʃ/. We consider the task to be passed if a correct rhyme is within the top ten tokens the model predicts as the next token. Although our

	Coherent R.V.	Incoherent R.V.
Pass	55%	0%
Fail	25%	18%

sample size is quite small, the effect is clear. Rarely occur, if ever, is the task completed successfully when an incoherent result vector is produced. In every example we analyzed, having a coherent R.V. was a prerequisite for completing the task.

E. Result vectors cluster around the proposed vowel chart

The vowel chart we propose in Figure 5 aligns with phoneme vector geometries in PCA space as shown in Figure 1. Importantly, this geometry is also present among result vectors. We use the same PCA to transform the H13L12 result vectors from 2000 single-token single-vowel words. Figure 9 shows that up to linear transformation[4], result vectors cluster around the phoneme vectors corresponding to their vowels.

[4]The result vectors tend to have much smaller magnitude than our phoneme vectors due to differences in normalization, so we scale up the PCs of the result vectors by a factor of 25 and shift each component by +8.

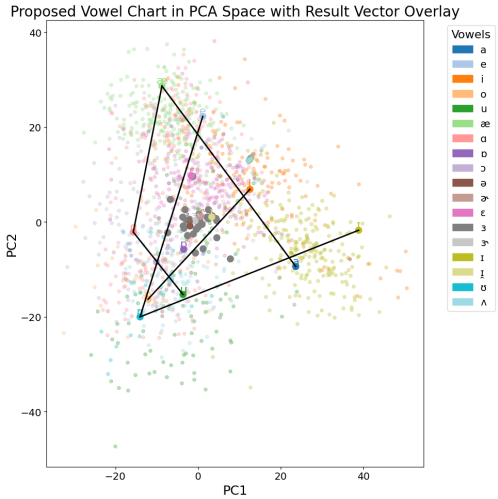

Figure 9: (High opacity) phoneme vectors corresponding to common English vowels in PCA space.
(Low opacity) H13L12 result vectors of single-token single-vowel words in the context of our rhyming task, scaled and shifted. The relative geometry of these result vector clusters matches the geometry of the proposed vowel chart in Figure 5.